BiasAperture — Demographic Fairness Audit Dossier

Diagnostic Demographic Bias Evaluation & Regulatory Traceability Report

AUDITED MODEL
FairFace ResNet-34 Multi-Task Classifier

EVALUATION DATASET
FairFace Benchmark Dataset (N=10954)

PROTECTED AXIS
Race

AUDIT DATE
2026-10-19 02:52:51 UTC

Diagnostic Scope & Non-Intervention Guarantee:

BiasAperture is engineered exclusively as an external diagnostic auditing, statistical significance testing, and regulatory traceability platform. It does **not** perform model retraining, in-processing debiasing, or synthetic image generation. In adherence to statistical integrity safeguards (NFR-003), demographic subgroups with fewer than 30 samples ($n < 30$) have point estimates suppressed to prevent unreliable compliance claims.

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SUBGROUPS WITH SIGNIFICANT DISPARITY (P < 0.05)

Overall (Table 1)
WORST DISPARITY COHORT (28 METRIC FAILURES)

DIR = 0.713
WORST HEADLINE METRIC (OVERALL)

0
SUBGROUPS FLAGGED N < 30 (NFR-003)

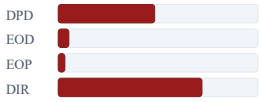
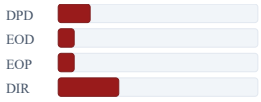
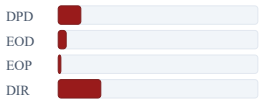
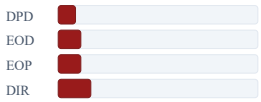
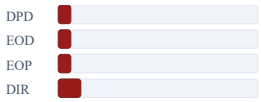
1. Headline Fairness Metrics (Core Four — Cross-Group Overview)

Summary metrics evaluated across all eligible demographic strata (n ≥ 30 per NFR-003). Accompanied by 95% stratified BCa bootstrap confidence intervals and Pearson's χ^2 test of independence. Plotted on a calibrated [0.0, 1.0] domain with null-point reference and threshold lines.

Metric Name	Point Estimate	95% Bootstrap CI	CI Band vs. Null	Significance (p-value)	Regulatory Clause
Demographic Parity Difference	0.192	[0.156, 0.222]		p=0.0000 **p < 0.05**	EU AI Act Art. 10(2)(f) / NIST Measure 2.11
Equalized Odds Difference	0.053	[0.035, 0.068]		p=0.0000 **p < 0.05**	EU AI Act Art. 10(2)(g) / NIST Measure 2.11
Equal Opportunity Difference	0.053	[0.034, 0.069]		p=0.0000 **p < 0.05**	EU AI Act Art. 10(3) / NIST Measure 2.11
Disparate Impact Ratio	0.713	[0.676, 0.760]		p=0.0000 **p < 0.05**	EEOC 4/5ths Rule / EU AI Act Annex IV §2(g)

2. Per-Subgroup Disparity Matrix

Detailed stratification across individual demographic groups, ordered by largest deviation from the null (no-disparity) point, most significant first. Subgroups below the NFR-003 sample threshold (n < 30) are strictly flagged, uncomputed to prevent small-sample estimation artifacts, and summarised separately below.

Subgroup	Sample Size (n)	DPD	EOD (Max-Gap)	EOP	DIR (Symmetric)	Deviation from Null (red = p < 0.05)	Data Integrity Status
Middle Eastern	1209	0.145	0.016	0.010	0.784		Audited (Valid)
Latino_Hispanic	1623	0.048	0.024	0.024	0.909		Audited (Valid)
Indian	1516	0.034	0.012	0.004	0.936		Audited (Valid)
Black	1556	0.026	0.034	0.034	0.951		Audited (Valid)
White	2085	0.019	0.019	0.019	0.966		Audited (Valid)
East Asian	1550	0.017	0.018	0.018	0.968		Audited (Valid)
Southeast Asian	1415	0.013	0.009	0.000	0.975		Audited (Valid)

3. Governance Documentation & Intended Use

> **EU AI Act (Regulation EU 2024/1689) Article 10 / Annex IV Alignment:** This audit dossier provides verifiable diagnostic documentation of demographic performance disparities. It does not perform model retraining or in-situ weight modifications.

Model Description: ImageNet-pretrained ResNet-34 terminating in an 18-unit linear multi-task head sliced for race (7), gender (2), and age (9) classification.

Backend configuration: dual. Both backend executions completed.

Cross-backend agreement checks point estimates only; bootstrap, hypothesis tests, and reporting share implementation. Consult execution logs for backend failures and divergences.